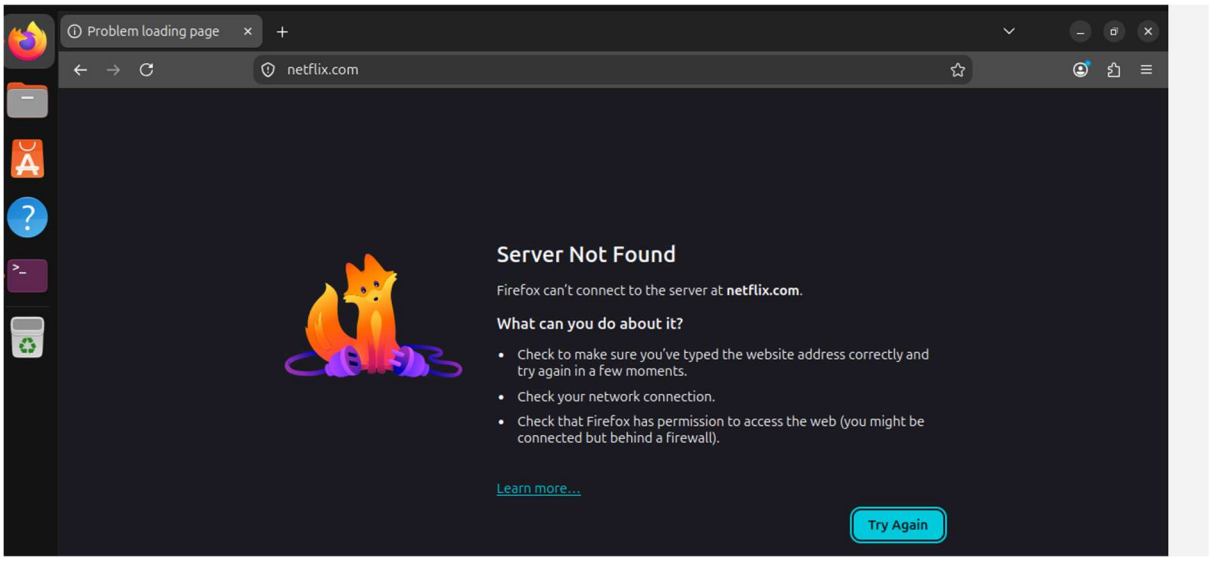
Field	Entry
Machine	Ubuntu26
Date	15 OCT 2026
Ticket	1 of 2

1.Symptom

Attempting to open netflix.com in Firefox resulted in long wait before an error appeared



2.Hypothesis-what you think is wrong, and why

The page is slow and failing to load rather than the site being obviously down, and the machine's network itself is running normally. That points at this machine's network path (its address, prefix or route) rather than the websites. I will check my on settings first, then test by Ip address before testing names.

3.Tests and Evidence-commands and output, include tests that passed.

Test 1: Is the interface up with an IPv4 address?

Command: ip a

Output:

```
cut@UbuntuVM:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 4000 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:e8:f6:0d brd ff:ff:ff:ff:ff:ff
    altname enx080027e8f60d
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 86233sec preferred_lft 86233sec
    inet6 fd17:625c:f037:2:a00:27ff:fee8:f60d/64 scope global dynamic noprefixroute
       valid_lft 86235sec preferred_lft 14235sec
    inet6 fe80::a00:27ff:fee8:f60d/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Result: pass

Meaning: enp0s3 is Up with 10.0.2.15/24 address.

Test 2: Is there a default route, and is the gateway in my subnet?

Command: ip r

Output:

```
       valid_lft forever preferred_lft forever
cut@UbuntuVM:~$ ip r
default via 10.0.2.99 dev enp0s3 proto static metric 20050
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 20100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
cut@UbuntuVM:~$
```

Result: Pass but suspicious

Meaning: Two default routes

Test 3: Can I reach my gateway?

Command: ping -c 2 10.0.2.99

Output:

```
cut@UbuntuVM:~$ ping -c 2 10.0.2.99
PING 10.0.2.99 (10.0.2.99) 56(84) bytes of data.
From 10.0.2.15 icmp_seq=1 Destination Host Unreachable
From 10.0.2.15 icmp_seq=2 Destination Host Unreachable

--- 10.0.2.99 ping statistics ---
2 packets transmitted, 0 received, +2 errors, 100% packet loss, time 1054ms
pipe 2
```

Result: Fail

Meaning: Destination Host Unreachable, 100% loss

Test 4: Can I reach the internet by Ip address?

Command: ping -c 2 8.8.8.8 (and ping 1.1.1.1 if one of them fails)

Output:

```
cut@UbuntuVM:~$ ping -c 2 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
From 10.0.2.15 icmp_seq=1 Destination Host Unreachable
From 10.0.2.15 icmp_seq=2 Destination Host Unreachable

--- 8.8.8.8 ping statistics ---
2 packets transmitted, 0 received, +2 errors, 100% packet loss, time 1000ms
```

Result: Fail

Meaning: Destination Host Unreachable, 100% loss.

Test 5: Can I resolve a name?

Command: ping -c 2 google.com

Output:

```
    cache
cut@UbuntuVM:~$ ping -c 2 google.com
ping: google.com: Name or service not known
cut@UbuntuVM:~$
```

Result: Fail

Meaning: Name or service not known

Test 6: Can I reach the DHCP gateway?

Command: ping -c 2 10.0.2.2.

Output:

```
pipe 2
cut@UbuntuVM:~$ ping -c 2 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=64 time=0.607 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=64 time=0.603 ms

--- 10.0.2.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 0.603/0.605/0.607/0.002 ms
```

Pass. The local network and that gateway work.

TEST7: Did anything answer for 10.0.2.99?

Command: ip neigh.

Result: INCOMPLETE,

Output:

```
cut@UbuntuVM:~$ ip neigh
10.0.2.99 dev enp0s3 INCOMPLETE
10.0.2.2 dev enp0s3 lladdr 52:54:00:12:35:00 REACHABLE
fd17:625c:f037:2::3 dev enp0s3 lladdr 52:54:00:12:35:00 router DELAY
fe80::2 dev enp0s3 lladdr 52:54:00:12:35:00 router STALE
```

Meaning: 10.0.2.99 … INCOMPLETE, so nothing answers. 10.0.2.2 is reachable.

Test 8: Which route is really used for 8.8.8.8?

Command: ip route get 8.8.8.8.

Output:

```
Command "table" is unknown, try "ip route help".
cut@UbuntuVM:~$ ip route get 8.8.8.8
8.8.8.8 via 10.0.2.99 dev enp0s3 src 10.0.2.15 uid 1000
    cache
cut@UbuntuVM:~$
```

Results: Fail via 10.0.2.99, the dead gateway

4.Root Cause-the actual cause, stated separately from symptoms

The machine has two default routes, and the preferred one, via 10.0.2.99 (metric 200500, points at gateway where nothing answers. Outside traffic is therefore sent to a dead end, while working gateway 10.0.2.2 (metric 20100) is never used. The interface, address and prefix are fine.

5.Fix-what you changed

1.Removed the bad route. (My first attempt had a typo, 10.0.2.999, and the system rejected it as an invalid address. Corrected it and re-ran.)

```
cut@UbuntuVM:~$ sudo ip route del default via 10.0.2.999
[sudo: authenticate] Password:
Error: any valid address is expected rather than "10.0.2.999".
cut@UbuntuVM:~$ sudo ip route del default via 10.0.2.99
```

2.Cleared the static gateway and routes from the saved connection profile, so the problem would not return after a reboot, then reactivated the connection:

```
cut@UbuntuVM:~$ sudo nmcli con mod netplan-enp0s3 ipv4.gateway '' ipv4.routes ''
cut@UbuntuVM:~$ sudo nmcli dev connect enp0s3
Device 'enp0s3' successfully activated with '1eef7e45-3b9d-3043-bee3-fc5925c90273'.
```

6.Verification-the same test now passing

ip route get 8.8.8.8 now goes via 10.0.2.2, and ping -c 2 8.8.8.8 succeeds with 0% packet loss:

```
cut@UbuntuVM:~$ ip route get 8.8.8.8
8.8.8.8 via 10.0.2.2 dev enp0s3 src 10.0.2.15 uid 1000
    cache
cut@UbuntuVM:~$ ping -c 2 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=7.40 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=7.29 ms

--- 8.8.8.8 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 7.293/7.345/7.398/0.052 ms
```

Field	Entry
Machine	Ubuntu26
Date	15 October 2026
Ticket	2 of 2

1.Symptom

The Cut website could not be opened in Firefox**.**

2.Hypothesis

Pings by IP address work but the name does not resolve, so the network path is intact, and the fault is in name resolution. I will find which DNS server this machine uses and whether it answers, then check the local files that affect name lookups.

3.Tests and Evidence

Test 1: Is there a single, valid default route?

Command: ip r

Output:

```
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
```

Result: Pass

Meaning: One default route via 10.0.2.2

Test 5: Can I resolve a name?

Command: ping -c 2 google.com

Output:

```
cut@UbuntuVM:~$ ping -c 2 google.com
ping: google.com: Name or service not known
```

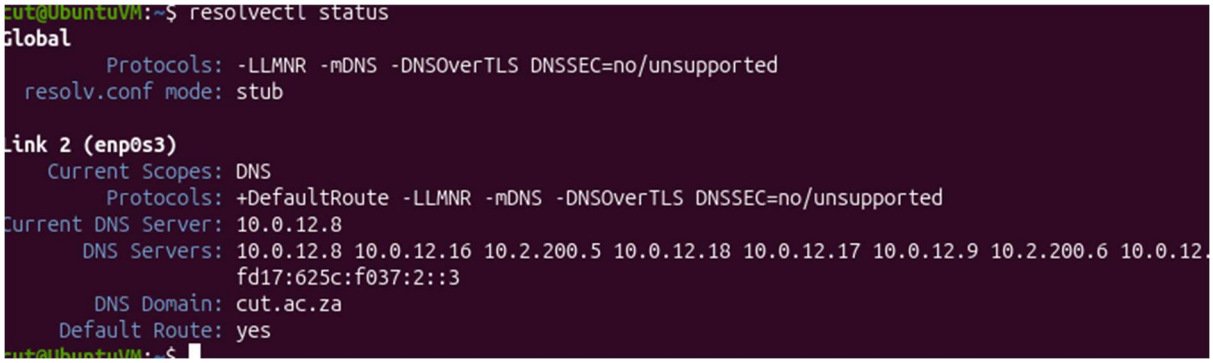
Result: Fail

Meaning: Name or service not known

Test 6: Which DNS server am I using, and does it answer?

Command: ping -c 2 google.com

Output:



Meaning: Current DNS server 10.0.12.8, which does not answer.

4.Root Cause

Dead DNS server: The configured DNS server 10.0.12.8does not answer queries, so no name can be resolved. Traffic by IP address is unaffected because it never uses DNS.

5.Fix

Planned approach: I was going to point the connection at a working DNS server, then reconnect.

6.Verification

Planned approach: re-run ping -c 2 google.com and resolvectl status and attach screenshots.